Quantum Computing, VQE and Results for Molecules

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October 29, 2025

1 Introduction

Quantum computing leverages quantum mechanical principles to solve problems in chemistry and physics with significant speed-up over classical algorithms for certain tasks. This project explores quantum algorithms aimed at finding ground-state energies of small molecules using variational methods, focusing on implementation via Qiskit and analysis of convergence, accuracy, and practical considerations.

2 Methods

2.1 Molecules Studied

The Jupyter notebook implements ground-state energy calculations for the following molecules:

- Lithium Hydride (LiH)
- Hydrogen (H_2)
- Water (H_2O)

Atomic positions are specified by 3D coordinates; various Hamiltonians are generated using PySCF and Qiskit Nature.

2.2 Hamiltonian Mapping and Ansätze

Hamiltonians derived via second quantization are mapped to qubit operators using the Jordan-Wigner transformation. Two types of ansätze are used for the Variational Quantum Eigensolver (VQE):

- **UCCSD** (Unitary Coupled-Cluster Singles and Doubles)
- **TwoLocal** (Hardware-efficient ansatz)

Active space reduction is utilized where appropriate to simplify the Hamiltonian and reduce computational complexity.

2.3 VQE and Optimization

The VQE method seeks the minimum eigenvalue of the mapped Hamiltonian using classical optimizers. `L-BFGS-B` is the primary optimizer, and all results are simulated in an ideal (noise-free) setting using Qiskit Aer primitive Estimator.

3 Problems, Solutions, and Results

Each molecule study follows a systematic procedure:

1. Define molecular geometry and run PySCFDriver to generate electronic problem and Hamiltonian.
2. If needed, apply active space transformation for electron/orbital selection.
3. Map second quantized Hamiltonian to qubit operators via Jordan-Wigner (or Bravyi-Kitaev, optional).
4. Build the chosen ansatz (UCCSD or TwoLocal).
5. Run VQE using L-BFGS-B optimizer, storing energy at each iteration.
6. Compare VQE result with the exact diagonalization (NumPyMinimumEigensolver).

3.1 Problem 1: LiH (Lithium Hydride)

Geometry: Li at (0,0,0); H at (0,0,1.6)
Active Space: 2 electrons, 3 orbitals
Ansätze: UCCSD, TwoLocal
Results:(Rounded)

- VQE energy (UCCSD): -7.8627 Ha
- VQE energy (TwoLocal): -7.8619 Ha
- *Exact energy:* -7.8629 Ha
- Number of qubits: 6
- Error (UCCSD): 2.30×10^{-4} Ha (0.14 kcal/mol)
- Error (TwoLocal): 1.05×10^{-3} Ha (0.66 kcal/mol)

3.2 Problem 2: H_2 (Hydrogen)

Geometry: H at (0,0,0) and (0,0,0.74)
Ansätze: UCCSD, TwoLocal
Results:(Rounded)

- VQE energy (UCCSD): -1.13728 Ha
- VQE energy (TwoLocal): -1.13728 Ha
- *Exact energy:* -1.13728 Ha
- Number of qubits: 6 (reported; actual qubit count may be 4)
- Error (UCCSD): 4.43×10^{-13} Ha (~ 0 kcal/mol)
- Error (TwoLocal): 1.42×10^{-10} Ha (~ 0 kcal/mol)

3.3 Problem 3: H_2O (Water)

Geometry: O at (0,0,0), H at (0,−0.757, 0.587), H at (0,0.757, 0.587)
Active Space: 2 electrons, 3 orbitals
Ansätze: UCCSD, TwoLocal
Results:(Rounded)

- VQE energy (UCCSD): −74.9646 Ha
- VQE energy (TwoLocal): −74.9631 Ha
- *Exact energy:* −74.9646 Ha
- Number of qubits: 6
- Error (UCCSD): 1.45×10^{-12} Ha (~ 0 kcal/mol)
- Error (TwoLocal): 1.57×10^{-3} Ha (0.99 kcal/mol)

4 Analysis and Discussion

The UCCSD ansatz consistently achieves near-exact ground state energies in all cases—its performance is significantly better than hardware-efficient ansatz (TwoLocal) for molecules with more correlation. The TwoLocal ansatz provides good results but exhibits higher errors, particularly for larger systems.

The active space approach improves computational efficiency while maintaining accuracy for small molecules. The Jordan-Wigner mapping is suitable for these systems; alternative mappings may be considered for systems with more qubits or different symmetries.

5 Limitations

The calculations are performed in an idealized noise-free simulation environment. Real hardware would introduce noise and decoherence, impacting accuracy. Further, the small molecules studied are well within the reach of classical methods, and the scaling advantage of quantum methods emerges for larger, more complex systems. The chosen active space and ansatz may not generalize to strongly correlated molecules.

6 Conclusion

This notebook demonstrates practical implementation of VQE for electronic structure problems, verifies accuracy against exact solutions, and discusses efficiency trade-offs of different ansätze. The approach is extensible to other molecules, with active spaces and mapping strategies tuned for hardware and problem size.

7 References

- Qiskit Documentation: `https://qiskit.org/documentation/`

- Qiskit Nature: `https://qiskit.org/ecosystem/nature/`
- PySCF: `https://pyscf.org/`
- S. McArdle, S. Endo, A. Aspuru-Guzik, et al. "Quantum computational chemistry." Reviews of Modern Physics **92**, 015003 (2020).
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